25901638 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Core Course (CC) - MG2CCRBCA102 - OPERATING SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. List any two process control system calls. [U] / [CO1]
2. Name any two CPU scheduling algorithms that use preemptive scheduling [An] / [CO2]
3. Explain different approaches for deadlock detection. [U] / [CO3]
4. How can a process execute its critical section? [U] / [CO3]
5. How does shared pages help in memory management? [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Discuss Dual Mode operation in operating system. [U] / [CO1]

7. Write notes on Layered operating system structure. [K] / [CO1]

8. Distinguish a process and a job and explain the queues and schedulers related to them. [U] / [CO2]

9. Explain the concept of threads. [U] / [CO2]

10. Define a semaphore. Explain its basic operations and how it is used in process synchronization. [U] / [CO3]

11. Differentiate between shared memory and message passing in IPC. [U] / [CO3]

12. Explain the concept of swapping with a diagram. [U] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. What is the purpose of context switching in process scheduling? [An] / [CO2]

14. Explain deadlock prevention techniques in detail [U] / [CO3]

15. Describe the segmentation technique in memory management . [A] / [CO4]

[15x2 = 30]